**Figure 11.6 The assistant processing the goal and outputting the results**

To demonstrate the effectiveness of the OpenAI Assistant's planner, we added another task, summarizing each page, to the goal. The inserted task didn't have a function/tool, but the assistant was savvy enough to use its ability to summarize the content. You can see the output of what the assistant produced by opening the `[root folder]assistants_working_folder/Wikipedia_{topic}.txt` file and reviewing the contents. Now that we understand how LLMs function without planners and planning, we can move on to creating our planners in the next section.

# 11.3 Building a sequential planner

LLM tools such as LangChain and Semantic Kernel (SK) have many planners using various strategies. However, writing our planner is relatively easy, and Nexus also supports a plugin-style interface allowing you to add other planners from tools such as LangChain and SK, or your derivatives.

Planners may sound complicated, but they are easily implemented through prompt engineering strategies that incorporate planning and reasoning. In chapter 10, we covered the basics of reasoning and deriving plans, and now we can put those skills to good use.

Listing 11.3 shows a sequential planner derived from the SK, which is extended to incorporate iteration. Prompt annotation planners like those shown in the listing can be adapted to fit specific needs or be more general like those shown. This planner uses JSON, but planners could use any format an LLM understands, including code.